

List

- Mutable
- Allows duplicates
- Hold multiple data types

```
In [1]: x = ["yasir", 20, 7.7]
        print(x)
```

```
['yasir', 20, 7.7]
```

```
In [2]: print(x.insert(1,"arafat"))
```

```
None
```

```
In [3]: print(x)
```

```
['yasir', 'arafat', 20, 7.7]
```

```
In [4]: print(x.append("single"))
```

```
None
```

```
In [5]: print(x)
```

```
['yasir', 'arafat', 20, 7.7, 'single']
```

```
In [6]: print(x.remove("yasir"))
```

```
None
```

```
In [7]: print(x)
```

```
['arafat', 20, 7.7, 'single']
```

```
In [8]: print(x.pop(2))
```

```
7.7
```

```
In [9]: print(x.count(20))
```

```
1
```

```
In [10]: print(x.reverse())
         print(x)
```

```
None
```

```
['single', 20, 'arafat']
```

```
In [11]: y = [2,4,6,2,1]

         print(y.sort())
         print(y)
         print(y.reverse())
         print(y)
         print(y.copy())
```

```
None
[1, 2, 2, 4, 6]
None
[6, 4, 2, 2, 1]
[6, 4, 2, 2, 1]
```

```
In [12]: a = [1,2,2,2,3,2]
         b = [5,5,6,6,7,7]
         print(a+b)

[1, 2, 2, 2, 3, 2, 5, 5, 6, 6, 7, 7]
```

```
In [13]: print(a*2)

[1, 2, 2, 2, 3, 2, 1, 2, 2, 2, 3, 2]
```

```
In [14]: print(list(set(a)))
         #remove all duplicates values

[1, 2, 3]
```

```
In [15]: print(list(set(b)))
         #remove all duplicates values

[5, 6, 7]
```

```
In [16]: z = [[1,2,3],[4,5,6]]
         print(z[1][2])

6
```

```
In [17]: # Create a list with integers from 1 to 5.

         numbers = list(range(1,6))
         print(numbers)

[1, 2, 3, 4, 5]
```

```
In [18]: l = [1,2,3]

         l[1] = 69

         print(l)

[1, 69, 3]
```

```
In [19]: # Check if a given element exists in a list.

         name = ["md", "yasir", "arafat"]

         if "yasir" in name:
             print("name is present")
         else:
             print("name does not exist")

name is present
```

```
In [20]: # Repeat a list 3 times using the * operator.

         l = [1,2,3]

         print(l*3)
```

```
[1, 2, 3, 1, 2, 3, 1, 2, 3]
```

```
In [21]: # Remove an element by value.
```

```
l = [1,99,"yasir",3]
```

```
l.remove("yasir")
```

```
print(l)
```

```
[1, 99, 3]
```

```
In [22]: # Remove an element by index.
```

```
l = [1,99,"yasir",3]
```

```
x = l.pop(2)
```

```
print(l)
```

```
[1, 99, 3]
```

```
In [23]: # Remove an element by index.
```

```
l = [1,99,3,"yasir"]
```

```
x = l.pop(-1)
```

```
print(x)
```

```
yasir
```

```
In [24]: # Sort a list in ascending order.
```

```
l = [1,4,2,6,3,1]
```

```
l.sort()
```

```
print(l)
```

```
[1, 1, 2, 3, 4, 6]
```

```
In [25]: # Sort a list in descending order.
```

```
l = [1,4,2,6,3,1]
```

```
l.sort(reverse=True)
```

```
print(l)
```

```
[6, 4, 3, 2, 1, 1]
```

```
In [26]: # Reverse a list without using reverse().
```

```
numbers = [1, 2, 3, 7, 5]
```

```
x = numbers[::-1]
```

```
print(x)
```

```
[5, 7, 3, 2, 1]
```

```
In [27]: # Find the sum of all elements in a numeric list.

numbers = [1, 2, 3, 7, 5]

print(sum(numbers))
```

18

```
In [28]: # Find the maximum and minimum in a numeric list.

numbers = [1, 2, 3, 7, 5]

print(max(numbers))

print(min(numbers))
```

7

1

```
In [29]: # Remove duplicates from a list.

t = [1,2,3,4,4,3,3]

print(list(set(t))) #first method

result = []      #SECOND METHOD
for num in t:
    if num not in result:
        result.append(num)

print(result)
```

[1, 2, 3, 4]

[1, 2, 3, 4]

In []: